Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Tobin, Brian W. Ph.D.		
Project Title:	Nutritional modulation of pancreatic endocrine function in microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Macon	**State:**	GA
Zip Code:	31207	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leeper-Woodford, Sandra (Mercer University School of Medicine) Uchakin, Peter (Mercer University School of Medicine) Lakey, Jonathan (University of Alberta) Smith, Scott (NASA-Johnson Space Center)		
Grant/Contract No.:	NCC 9-58-NPFR.002.04		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ground-based and in-flight investigations illustrate changes in insulin, glucose, and amino acid metabolism in space flight. These observations may relate to altered pancreatic endocrine function, which is insufficient to meet the needs of microgravity-induced insulin resistance, and altered amino acid metabolism. The changes observed include decreased glucose tolerance, increased circulating insulin, and increased reliance upon glucose in muscles. The metabolic milieu resembles an insulin resistant syndrome, accompanied by a compensatory increase in pancreatic insulin secretion. However, the increase in insulin secretion is insufficient to ameliorate muscle atrophy. The increased insulin secretion is well correlated to muscle atrophy in space flight. The influence of these changes upon the loss of muscle mass and general endocrine metabolic state are not well established, however. Countermeasures which could modulate insulin and glucagon secretion in a compensatory manner to overcome insulin resistance and promote amino acid uptake by peripheral musculature might decrease muscle atrophy and reduce injury following re-adaptation to unit gravity. <P> We hypothesize that human pancreatic islets of Langerhans have an increased requirement for amino acids in microgravity. We further hypothesize, that supplementation with specific additional amino acids will augment, enhance and normalize insulin secretion, when space flight paradigm stressors known to decrease insulin secretion, are applied. Our specific aims in this study are to: 1) assess the effect of a microgravity model cell culture on basal amino acid requirements and endocrine secretory function in human islets of Langerhans, and 2) determine human islet endocrine function while testing amino acid countermeasures in the microgravity model. <P> It is anticipated that these studies will further refine our understanding of human pancreatic amino acid requirements and endocrine regulation: phenomenon which may be limiting to extended-duration space flight missions. These studies will test countermeasures to augment pancreatic endocrine function, while considering both insulin and glucagon production in a way that will involve supplementation of diet with additional amino acids. These measures are ultimately aimed at improving space flight induced muscle atrophy, and ameliorating current re-adaptation constraints. <P>
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies are designed to further our understanding of human pancreatic amino acid requirements and endocrine function: phenomenon which are limiting to extended duration spaceflight and human pancreatic islet transplantation in diabetes therapies. These studies illustrate similarities between the effects of diabetes and spaceflight upon human metabolism in pancreatic islets of Langerhans. The investigations support the use of ground-based microgravity model systems to study the cellular physiology of pancreatic islets of Langerhans in diabetes research. These are the first studies to illustrate cellular aggregation af pancreatic islets of Langerhans using microgravity model systems. The cellular endocrine pancreas aggregates may prove useful to the development of co-cultured endrocrine cells, suitable for pancreatic islet transplantation, as a therapy for diabetes. In addition these studies are elucidating potentially important data which may be used in the managment of ICU myopathy, which is characterized by bedrest, cortisol excess, and muscle loss.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)